

A Phase-Walk-Based True Random Number Generator Exploiting Dual-Ring Phase Jitter Comparison

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Abstract—A True Random Number Generator (TRNG) is a critical security component that generates unpredictable random numbers based on physical entropy sources, widely used in cryptography, secure communications, and simulation. TRNGs leverage non-deterministic physical phenomena like circuit noise and jitter to ensure randomness. To enhance the security of traditional ring oscillator (RO), multi-phase quantization sampling, logical circuit mapping is realized by using Time-to-Digital Converter (TDC) to improve the performance and security of the entropy source. To meet the demands of high performance and low overhead, this brief proposes a phase-walk based TRNG. By comparing two pairs of ROs and utilizing random phase jitter, the proposed TRNG achieves 600 Mbps throughput on Artix-7, demonstrating unique advantages in resource efficiency and performance.

Index Terms—TRNGs, FPGA, phase walk, time-to-digital converter, high-throughput.

I. INTRODUCTION

TRUE Random Number Generators (TRNGs) play a critical role in information security, cryptography, hardware security, and modern computing systems. TRNG rely on physical entropy sources, such as electronic noise, clock jitter to generate unpredictable random numbers, offering higher security. They are widely used in key generation, security protocols, IoT device authentication, blockchain, and trusted computing.

In recent years, with the rapid development of quantum computing and artificial intelligence, the security of traditional encryption algorithms faces severe challenges, driving increasing demand for TRNGs with high entropy, low latency, and

low cost. However, TRNG design still confronts challenges in entropy source stability, area overhead, and energy efficiency.

Phase-based TRNG leverage timing uncertainties such as clock jitter [1], this design employs logic gates to perform time-threshold processing on the accumulated jitter noise, and a unique delay capture circuit is proposed to convert the delay signal into a control signal. Meanwhile, there are also designs that utilize signal phase noise [2], or delay lines as entropy sources [3], offering unique advantages, but require precise clock management imperfect and jitter or phase differences can introduce systematic biases that degrade randomness. In 2024, Fabio Frustaci proposed a structure for implementing oscillating units using Digital Signal Processor (DSP) [4], DSP is used instead of ROs, trading off portability for reduced resource consumption. Peng proposes a TRNG based on the metastability of shift registers [5] and adopts a metastable randomness extraction method to reduce the deviation of metastable outputs. In addition, there is another method that constructs iterative circuits through algorithmic mapping to generate random numbers [6], and multi-channel random number outputs are employed to improve the throughput.

Oscillator jitter and phase difference amplification are physically unpredictable random phenomena, capable of stably providing high-quality entropy sources. Typically, TRNG based on delay lines and clock phase sampling consume fewer resources, making them more suitable for low-power, small-area system designs. In 2022, Yao proposed a multi-phase sampling-based TRNG [7], fine-grained sampling is performed using clocks of different frequencies to achieve a multi-phase sampling effect. When a slight jitter occurs in the oscillation signal, XORing it with a stable signal enables rapid capture of phase variations, and this fast-response characteristic can be leveraged to implement a high-throughput TRNG architecture [8].

The higher the order of the RO, the greater the total jitter standard deviation σ . Jitter propagates and accumulates in the RO [9], manifesting as unpredictable phase noise in the frequency domain. Phase noise is more pronounced in high-frequency ROs because a higher frequency implies a shorter period, making phase more sensitive to timing jitter. However, as frequency increases, the RO's order decreases, leading to reduced jitter and degrade the quality of the entropy source.

Therefore, current RO-based TRNG designs must balance throughput and entropy quality. However, Time-to-Digital

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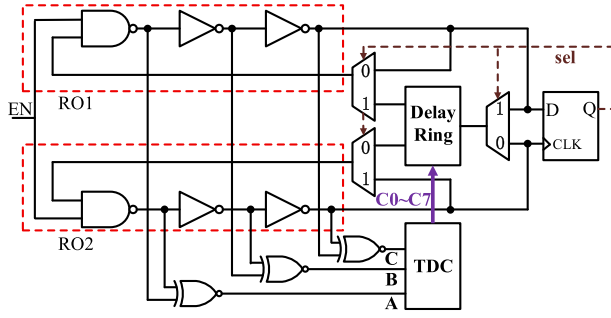


Fig. 1. Phase walk timing.

Converter (TDC) technology can effectively quantify minute jitter [10], adapting to diverse design scenarios. By combining TDC with phase jitter, the degradation in randomness caused by reduced Ring Oscillator order can be mitigated, enabling simultaneous improvements in both throughput and entropy.

Through the above research on oscillation loops and TDC, the random phenomena generated by oscillations can be digitally converted to achieve the visualization of randomness. Additionally, such random jitter is fed back in an alternative manner to influence the oscillation process.

II. PROPOSED HYBRID DUAL-RING STRUCTURE

Drawing on the current well-established methodologies, we propose a phase-walk based TRNG architecture based on the “phase comparison + TDC” scheme, as shown in Fig. 1. The structure mainly consists of two parts: an oscillation unit, and a delay system.

The oscillation unit is composed of two third-order RO rings and the TDC module, ROs are controlled by an input signal EN, ensuring both excellent frequency performance and cumulative phase noise generation. After each inverter stage, a tap signal undergoes an XNOR, with the result input to TDC module to generate control signals C0-C6.

The delay system comprises the arbitration system and the delay ring. The D flip-flops and MUX connected to the two ROs are responsible for arbitration, they randomly select one of the ROs for phase walking according to the output of sel, where the walking amplitude and the process of signal inversion are randomly regulated by C0-C6. Finally, the signals from the two ROs are XORed to generate the random number.

A. Delay Ring Structure

The delay loop is constructed using controllable XOR gates, with its specific structure shown in the Fig. 2. The overall architecture is composed of a combination of XOR gates and MUXs. This is because XOR gates are generally regarded as having dynamic switching characteristics that allow them to function as inverters and buffers, which can enhance randomness and provide stable time delays [11]. Meanwhile, MUXs can flexibly adjust the length of the delay chain according to the variation of the select signal.

The phase difference of the extracted signals is abstracted as a binary number “ABC”, which reflects the degree of

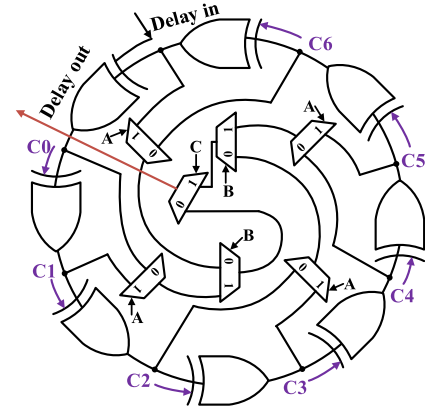


Fig. 2. Delay ring structure.

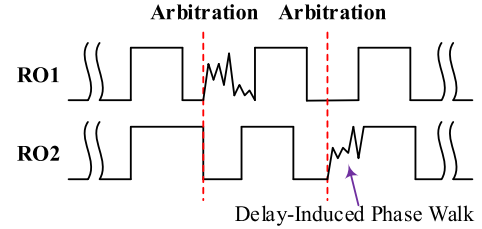


Fig. 3. Phase walk timing.

difference in jitter and phase walking between the two ROs. “ABC” determines multiple routing combinations. Each case from 000 to 111 corresponds to a designed control signal:

- 1) When “ABC” = 100, the walking signal must traverse 5 XOR gates before output.
- 2) When “ABC” = 111, the walking signal must pass through all XOR gates to output.

The other input of each XOR gate serves as the configuration terminal, where the configuration signals are mapped from TDC module. These configuration signals will form a random number of inverter-buffer combinations using the XOR gates. At this point, when jitter occurs in the RO, it will cause rapid variations in ABC. The XOR gates traversed will change accordingly, which not only alters the phase of the oscillation ring but also triggers signal inversion. The resulting effects are illustrated in Fig. 3.

B. The TDC Module and Phase-Walking Operation

“ABC” passes through a Linear Feedback Shift Register (LFSR) formed by 7 D flip-flops to generate a 7-bit control signal. The lower bits inject ABC into the LFSR via half-adder operations to form an entropy injection layer. The higher bits form a nonlinear transformation layer through feedback and permutation, as detailed in the formula (1).

$$\begin{cases} C0 = A \oplus C6 \\ C1 = B \oplus C0 \\ C2 = C \oplus C1 \\ C3 = C2 \oplus C6 \\ C4 = C0 \oplus C1 \oplus C3 \\ C5 = C1 \oplus C2 \oplus C4 \\ C6 = C1 \oplus C5 \end{cases} \quad (1)$$

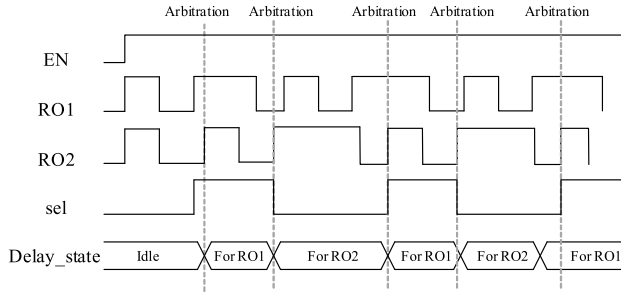


Fig. 4. Delay selection operation.

Therefore, the control signals C0–C6 are the comprehensive results of temporal memory, jitter, phase differentiation, and nonlinear transformation.

The operation of implementing random selection is illustrated in Fig. 4. Whenever the output waveforms of RO1 and RO2 connected to the arbiter change, the RO whose rising edge arrives at the arbiter first is preferentially selected for phase walking. The arbitrated output sel acts on the MUX in Fig. 1, and the selected RO is routed through the delay ring controlled by ABC to complete one cycle of “phase walking”.

III. PHASE WANDERING EFFECT

The phase jitter of the RO is the fundamental source of entropy. The oscillation period of each RO is non-ideal. We model the phase jitter of the RO within the k -th cycle as a random variable $J_K \sim N(0, \sigma^2)$.

At the initial moment ($t = 0$), RO₁ and RO₂ are enabled simultaneously. Due to the manufacturing process, even with a fully symmetric design, deviations exist between the initial phases and natural frequencies of the two ROs. Therefore, at any arbitrary time t , the absolute phase difference $\Delta\Phi(t)$ between RO1 and RO2 can be expressed as:

$$\Delta\Phi(t) = 2\pi \cdot \Delta f \cdot t + \Delta\Phi_0 + \Phi_{jitter}(t) \quad (2)$$

where $\Delta f = f_{RO1} - f_{RO2}$, $\Delta\Phi_0$ denotes the initial phase difference, $\Phi_{jitter}(t)$ represents the time-accumulated random jitter difference.

Let the arrival times of the n -th rising edge of RO1 and RO2 be $T_1[n]$ and $T_2[n]$, respectively. Their time difference $\Delta T[n] = T_1[n] - T_2[n]$ is a random variable. Since the jitter J_K of each cycle follows a normal distribution, the time difference $\Delta T[n]$ approximately obeys a normal distribution after accumulation over multiple cycles:

$$\Delta T[n] \sim N(\mu_{\Delta T}, \sigma_{\Delta T}^2) \quad (3)$$

$\mu_{\Delta T}$ is the mean value of the systematic deviation jointly determined by the frequency difference Δf and the initial phase difference $\Delta\Phi_0$. $\sigma_{\Delta T}^2$ is the variance of the time difference, which is related to the jitter variances of the two ROs. Assuming that the jitters of RO1 and RO2 are mutually independent, then $\sigma_{\Delta T}^2 = 2\sigma^2$.

Since the jitter of the RO after phase variation follows a normal distribution, the intermediate nodes should also exhibit favorable randomness. Taking A as an example. The probability that it is '1' at a certain sampling moment, denoted as $P(A) = 1$, is equal to the probability that the output signals

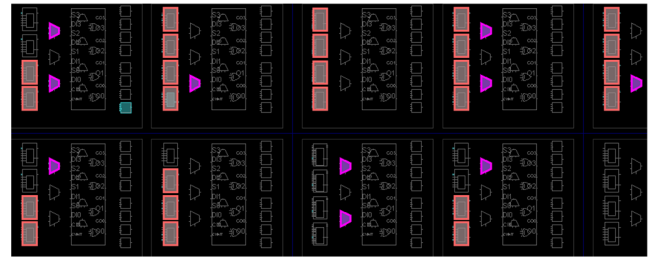


Fig. 5. Implementation with automated placement and routing.

TABLE I
INIST SP800-22 TEST

NIST SP800-22	Artix7		PYNQ	
	P-value	Prop	P-value	Prop
App-Entropy	0.213309	99/100	0.10879	99/100
Block-Freq	0.759756	97/100	0.699313	98/100
Cumsum*	0.395462	99/100	0.175905	99/100
FFT	0.437274	100/100	0.108791	99/100
Frequency	0.23681	99/100	0.595549	99/100
Line-Complex	0.350485	98/100	0.090936	98/100
Long-Run	0.616305	100/100	0.924076	99/100
Nonoverlapping*	0.546288	100/100	0.506977	99/100
Overlapping	0.911413	100/100	0.494392	99/100
Rand-Excur*	0.102172	14/14	0.290895	12 /12
Rand-Variant*	0.182158	14/14	0.254391	12 /12
Rank	0.006661	98/100	0.455937	99/100
Runs	0.851383	100/100	0.213309	97/100
Serial*	0.667734	100/100	0.470468	99/100
Universal	0.213309	100/100	0.213309	100/100

of the first stage of RO1 and RO2 are not equal at that moment. This is equivalent to the probability that the absolute value of the normalized phase difference $\Delta\Phi_A$ at the sampling point can cause a phase difference amplitude.

Considering the uncertainty near the threshold voltage, a more general model expresses this probability using the cumulative distribution function (CDF) of the normal distribution. Assuming that the output has uncertainty when the time difference falls within the interval $[-t_h, t_h]$, we have:

$$P(A = 1) = 1 - \int_{-t_h}^{t_h} f_{\Delta T_A}(x) dx \quad (4)$$

$$P(A = 1) = 1 - \left[\Phi\left(\frac{t_h - \mu_{\Delta T_A}}{\sigma_{\Delta T_A}}\right) - \Phi\left(\frac{-t_h - \mu_{\Delta T_A}}{\sigma_{\Delta T_A}}\right) \right] \quad (5)$$

If the systematic deviation $\mu_{\Delta T_A}$ is zero and the uncertainty window t_h is small relative to $\sigma_{\Delta T_A}$, then $P(A = 1) \approx 0.5$. However, $\mu_{\Delta T_A}$ is usually non-zero, which causes static deviations in signals A, B, and C. Therefore, when ABC is combined, three incompletely identical node deviations are generated, which introduces nonlinearity and reduces correlation.

IV. EXPERIMENTAL RESULTS

The proposed structure XORs the waveforms of RO1 and RO2 to produce the output, sampled by a 600 MHz clock. The detailed post-implementation layout is shown in Fig. 5.

A. NIST SP 800-22 Test

The values in the table with * are the results of averaging multiple values.

NIST SP 800-22 is one of the most widely used RNG test suites, comprising 15 test items to evaluate TRNG performance and distinguish true random number generators from

TABLE II
AIS-31

AIS-31	Rate	Result
T0	1/1	Pass
T1-T5	257/257	Pass
	$ P(1) - 0.5 = 2.9E-4$	Pass
T6	$p(01) = 0.50132; p(11) = 0.50173$	Pass
	$ p_{-}(01) - p_{-}(11) = 4.1E-4$	Pass
T7	$T[0] = 1.41; T[1] = 1.85$	Pass
	$T[2] = 1.35; T[3] = 1.98$	Pass
T8	H I = 8	Pass

TABLE III
NIST 90B

NIST 90B	P(max)	Min-entropy
MCV	0.501993	0.994261
Collision	0.529297	0.917851
Markov	3.97E-39	0.996611
Compression	0.5	1
t-Tuple	0.52065	0.941613
LRS	0.524149	0.931952
MultiMCW	0.501235	0.996442
Lag Prediction	0.501077	0.996897
MultiMMC	0.500732	0.997889
LZ78Y	0.501274	0.99633

pseudo-random number generators. In this brief, we collected ten million data from the output for testing. The results as shown in TABLE I, confirm that all random sequences passed the NIST SP 800-22 tests.

B. NIST SP 800-90B and AIS-31 Test

To further evaluate the randomness quality of the proposed TRNG, we also tested the entropy source using AIS-31 and NIST SP 800-90B. Table II reports the results of the AIS-31 test suite, where the byte entropy of the T8 test reaches an exceptionally high value. TABLE III shows the minimum entropy estimation using non-iid test of NIST SP 800-90B. The results demonstrate that the TRNG can generate true random number sequences with high bit entropy, meeting the requirements for high-performance, high-entropy TRNG.

C. TESTU01 Test

TESTU01 is regarded as a more stringent randomness test capable of detecting more subtle random deviations. We input a 1-Gbit sequence into the Rabbit battery of TestU01-1.2.3, and the results are presented in Table IV. This test suite aims to efficiently detect weaknesses in TRNGs (e.g., LFSRs, linear congruential generators) based on linear operations. It is particularly adept at identifying structural flaws such as linear correlations and periodicity in random sequences at both the bit level and byte level [12]. The P-values in the test report all fall within the passing range [0.001, 0.9990], and all test items are successfully passed, demonstrating that the “authenticity” and “unpredictability” of the randomness of the proposed structure are extremely high.

TABLE IV
TESTU01

Statistical	P-value	P/F
smultin-Mul-Bit-Ove	0.07	Pass
snpair-Clo-Pai-Bit-Mat $t=(2,4)$	(0.52, 0.17)	Pass
svaria-App-Spa	0.44	Pass
scomp-Lin-Com	0.55	Pass
scomp-Lem-Ziv	0.22	Pass
sspectral-Fou-1	0.73	Pass
sspectral-Fou-3	0.09	Pass
sstring-Lon-Hea-Run	0.26	Pass
sstring-Ham_wei	0.67	Pass
sstring-Ham-Cor $L=(32,64,128)$	(0.04, 0.08, 0.50)	Pass
sstring-Ham-Ind $L=(16,32,64)$	(0.14, 0.83, 0.18)	Pass
sstring-Aut-Cor $d=(1,2)$	(0.78, 0.81)	Pass
sstring-Per-InS	0.76	Pass
sstring-Run	0.89	Pass
smarsa-Mat-Rank with 320×320	0.57	Pass
swalk Ran-Walk1(H,M,J,R,C)	(0.33, 0.16, 0.96, 0.03, 0.9915)	Pass

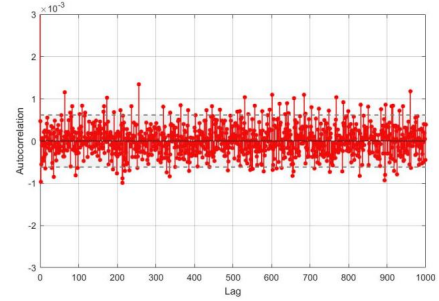


Fig. 6. Result of autocorrelation test.

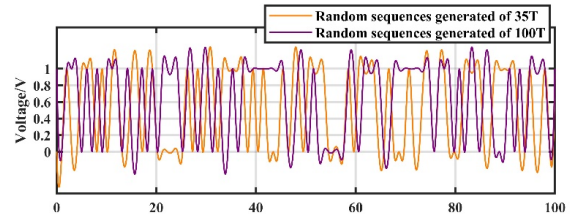


Fig. 7. Simulation of output sequences on different boards.

D. Autocorrelation Test

To ensure the statistical independence of the TRNG output, rigorous correlation testing must be performed on the generated sequences. The core objective is to verify whether observable statistical dependencies exist between any adjacent or non-adjacent bits. We extracted 1 million bits of random sequences and conducted autocorrelation tests.

As shown in the Fig. 6, the absolute value of the autocorrelation coefficient is below 0.003, confirming that the output random sequences are uncorrelated. Meanwhile, we tested the output signals on the xc7a35T and xc7a100T development boards. The results are shown in Fig. 7. As can be seen from the figure, the sequences generated on different boards are uncorrelated, as they exhibit different initial conditions and waveform behaviors at startup.

E. Reliability Experiments

The proposed TRNG was tested under different temperatures (20°–60°) and different voltages (0.9V–1.1V). The

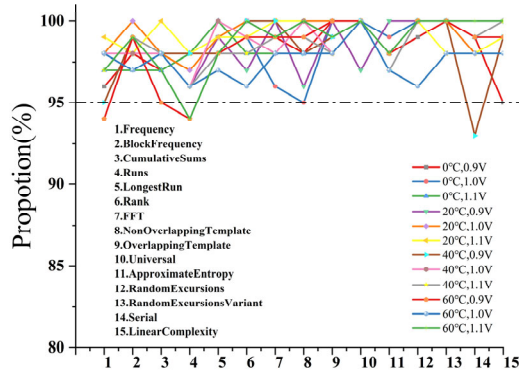


Fig. 8. Result of voltages and temperatures test.

TABLE V
COMPARISON OF THROUGHPUT AND HARDWARE RESOURCE
CONSUMPTION

Design	Platform	Resource	TP(Mbps)	Power(W)	Min-entropy or Byte entropy	Throughput Slices
[2]	Artix-7/Kintex-7	15LUTs 8DFFs 8MUXs	550	0.111	7.976(B)	91.6
[3]	ZYNQ-7000	21LUTs 36DFFs 17CARRY4	25	0.008	0.996(M)	0.86
[10]	spartan-6	53LUTs 32DFFs 62regs	280	0.036	0.996(M)	9
[11]	spartan-6/Virtex-6	13LUTs 4DFFs	350/500	110/112	0.936(M)	107.69
[13]	Virtex-6	24LUTs 2DFFs	290	3.678	0.946(M)	NA
[14]	Artix-7/Kintex-7	19LUTs 8DFFs	380	NA	0.990(M)	34.5
[15]	spartan-6	4LUTs 3DFFs	0.76	NA	7.998(B)	0.76
[16]	spartan-6	38LUTs 12DFFs	400	0.067	0.946(M)	33.3
This work	Artix-7	26LUTs 10DFF 10MUXs	600	0.182	0.994(M)	66.67

random sequences output by the circuit were collected and subjected to the NIST SP 800-22 test. We counted the pass rates of the 15 test items, and the specific results are shown in Fig. 8. The TRNG passed the tests in most cases and can still output high-quality random numbers in complex environments.

F. Comparison With the State-of-the-Art

The performance of this structure is compared with other advanced TRNGs in TABLE V. The test metrics for measuring entropy are different, for the rationality of comparison, we abbreviate the min-entropy as M and the byte entropy as B. While [3] and [15] demonstrate very low power consumption, [3] and [15] show poor throughput performance. Although the proposed configurable TDC circuit [10] can achieve higher throughput, it also requires more resource consumption. [2], [14], and [16] exhibit excellent performance. However, with the ever-increasing demand for performance, the proposed TRNG achieves a higher throughput in comparison, demonstrating its competitiveness in meeting practical requirements. It demonstrates competitiveness in designs with the same entropy source. Among these, a normalized metric for evaluating resource consumption and throughput is expressed as follows. $\frac{\text{Throughput}}{\text{Slices}}$. A higher value of this metric indicates a better throughput rate per unit state. [11] achieves better performance; however, this is enabled by employing dual output ports. In contrast, the proposed TRNG achieves considerable performance with a single output port while maintaining high entropy values, thus demonstrating a design philosophy that balances performance and resource utilization.

V. CONCLUSION

In this brief, we propose a novel TRNG and has implemented on Xilinx Artix-7. The proposed TRNG achieves a throughput of 600 Mbps. The phase-walk based TRNG architecture is implemented through the combination of phase delay comparison and TDC, digital control signals can be generated from the time-domain variations of the oscillation ring to regulate the amplitude of phase walking. Experimental results demonstrate that the designed TRNG exhibits excellent performance and stability, providing a novel design approach for architectures based on phase comparison and TDC.

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